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小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

Author: 留子也要好好吃面 • CC BY 4.0



摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房实验考察小苏打浓度与煮制时间对商品干制意大利面拉面化感官表现的共同影响。实验采用3 × 3处理矩阵，将2、4、6 g/L三种小苏打浓度与10、12、15分钟三种煮制时间组合，共形成九种处理。作者依据弹性、中心熟度、顺滑度、碱香、苦涩感及与拉面风格面条的整体相似度进行结构化描述，并给予10分制综合评分。4 g/L、15分钟处理得分最高（9.5/10）；4 g/L、12分钟得分8.5/10。2 g/L处理的变化总体较弱，而6 g/L处理随时间延长出现苦味、涩感、黏口及肥皂样气味。该结果支持将4 g/L、15分钟作为家庭厨房试制的实用起点，但由于研究为单人评价、未报告独立重复且缺少pH和仪器质构测量，结论应理解为探索性观察。

Abstract: This exploratory kitchen trial examined the combined effects of baking-soda concentration and cooking time on the ramen-style transformation of commercial dried spaghetti. A 3 × 3 matrix combined 2, 4, and 6 g/L baking soda with cooking times of 10, 12, and 15 minutes. The author documented elasticity, center doneness, smoothness, alkaline aroma, bitterness, and overall similarity to ramen-style noodles, then assigned a descriptive 10-point score. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes; 4 g/L for 12 minutes scored 8.5/10. Treatments at 2 g/L produced limited change, whereas prolonged cooking at 6 g/L produced bitterness, astringency, stickiness, and a soap-like odor. The findings support 4 g/L for 15 minutes as a practical starting condition, but remain exploratory because the trial involved one evaluator, undocumented replication, and no pH or instrumental texture measurements.

Keywords: baking soda; spaghetti; alkaline noodles; ramen substitute; sensory evaluation

1. Background / 背景

Fresh alkaline noodles are not consistently available in every market, whereas dried spaghetti is shelf-stable and widely available. Informal recipes often recommend adding an unspecified amount of baking soda, but concentration and time are rarely compared systematically. The study therefore compared nine concentration-time combinations and documented practical sensory differences.

2. Materials and Methods / 材料与方法

Each treatment used 1 L water. Baking-soda concentrations were 2, 4, and 6 g/L and cooking times were 10, 12, and 15 minutes. One evaluator described springiness, surface smoothness, center firmness, alkaline aroma, bitterness, and ramen-like similarity. Independent replication and randomization were not documented.



Figure 2. Commercial spaghetti and 4 g baking soda.

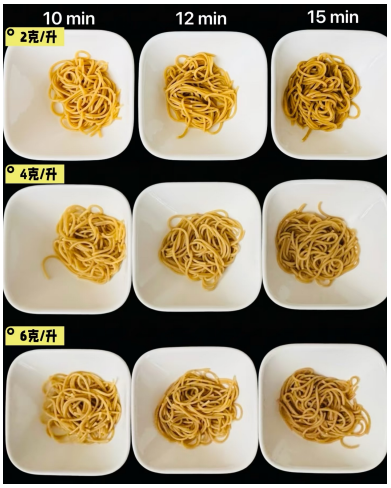


Figure 3. Nine treatments.

3. Results / 结果

g/L	10	12	15
2	5.0	6.0	6.5
4	7.0	8.5	9.5
6	6.0	5.0	3.5

At 2 g/L, scores improved only modestly with time. At 4 g/L, scores increased across time and peaked at 15 minutes. At 6 g/L, scores declined as bitterness, astringency, stickiness, and soap-like odor intensified.



Figure 4. Original score matrix.

4. Practical Procedure / 实用方法

Add 4 g baking soda to 1 L water, bring to a boil, add spaghetti, and cook for 15 minutes. A 12-minute endpoint may suit a firmer preference. Use a larger pot and monitor foaming.

5. Discussion / 讨论

The main pattern was not that more alkali produced a better noodle. A middle concentration had to be balanced with sufficient center softening. Photographic color changes should not be treated as instrumentally calibrated measurements.

6. Limitations / 局限性

Dry pasta mass, replication, randomization, blinding, pH, cooking loss, instrumental texture, and calibrated color were not documented. The results should be interpreted as practical exploratory observations.

7. Conclusion / 结论

Within this trial, 4 g/L for 15 minutes received the highest score. Future work should add standardized mass, replication, independent sensory evaluation, pH, texture, and cooking-loss measurements.

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五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性：一项探索性厨房比较

Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types Cooked in Alkaline Water: An Exploratory Kitchen Comparison

Author: 留子也要好好吃面 • CC BY 4.0



摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房比较在统一碱水条件下考察五种商品意大利面的尺寸变化、建议煮制时间和中式面条应用潜力。Angel Hair、Thin Spaghetti、Spaghetti、Linguine和Fettuccine均在1 L沸水加入4 g小苏打的条件下煮制。所报告的生面/熟面尺寸分别为1.2/1.8、1.4/2.1、1.8/2.8、3.1/4.6和4.6/6.8 mm；对应时间为5、9、15、17和18分钟。平均尺寸膨胀倍数为1.50。Angel Hair在3至8分钟内由1.5增加至1.9 mm，但5分钟被选为感官端点，说明最大膨胀并不等同于最佳口感。名义尺寸与建议时间呈描述性正相关（Pearson r = 0.82），但样本仅五种且混合了圆面直径与扁面宽度，因此结果只应作为实践参考。

Abstract: This exploratory kitchen comparison examined dimensional change, recommended cooking time, and potential Chinese-noodle applications for five commercial pasta formats under a standardized alkaline condition. Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine were cooked in 1 L boiling water containing 4 g baking soda. Reported dry/cooked dimensions were 1.2/1.8, 1.4/2.1, 1.8/2.8, 3.1/4.6, and 4.6/6.8 mm, with cooking times of 5, 9, 15, 17, and 18 minutes. Mean dimensional expansion was 1.50. Angel Hair increased from 1.5 to 1.9 mm from 3 to 8 minutes, although 5 minutes was selected as the sensory endpoint, indicating that maximum swelling did not equal optimal texture. Nominal size and recommended time showed a descriptive positive association (Pearson r = 0.82), but the five-item dataset mixed diameter and width measurements and should be treated as a practical reference only.

Keywords: alkaline cooking; caliper measurement; dimensional expansion; pasta format

1. Background / 背景

Commercial pasta formats differ in thickness, geometry, and conventional cooking instructions. The study compared five formats under a common alkaline condition to examine time, dimensional change, and possible Chinese-noodle applications.

2. Materials and Methods / 材料与方法

Products were independently purchased and cooked in 1 L boiling water with 4 g baking soda. Time was recorded at the author-defined al dente endpoint. Diameter was recorded for round pasta and width for Linguine and Fettuccine. Strand count, replication, calibration, and caliper pressure were not documented.



Figure 2. Compared pasta products.

3. Results / 结果

Type	Dry	Cooked	Ratio	Min
Angel Hair	1.2	1.8	1.50x	5
Thin Spag.	1.4	2.1	1.50x	9
Spaghetti	1.8	2.8	1.56x	15
Linguine*	3.1	4.6	1.48x	17
Fettuccine*	4.6	6.8	1.48x	18

* Flat-pasta values are widths; round-pasta values are diameters.

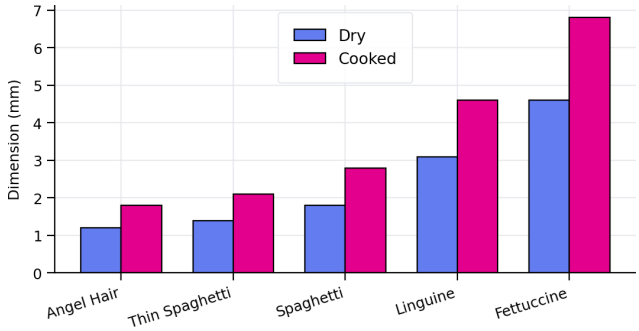


Figure 3. Dry and cooked dimensions.

Expansion ratios clustered from 1.48 to 1.56. Recommended time increased from 5 to 18 minutes. The descriptive correlation between nominal size and recommended time was r = 0.82, but the dataset is too small and heterogeneous for inferential claims.

4. Angel Hair Time Course / 时间进程

Angel Hair readings at 3, 4, 5, 6, 7, and 8 minutes were 1.5, 1.6, 1.7, 1.8, 1.8, and 1.9 mm. A firm center remained at 3 minutes, and 5 minutes was selected as the recommended sensory endpoint.

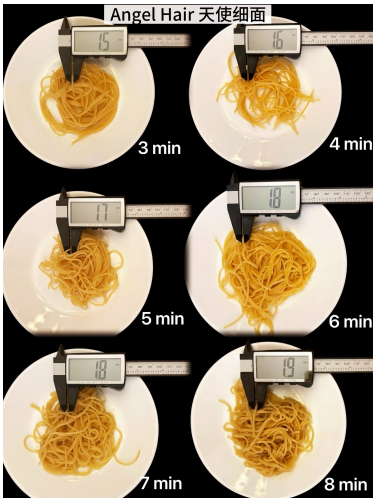


Figure 4. Angel Hair measurements.

5. Culinary Applications / 应用建议

Angel Hair and Thin Spaghetti suited fine-noodle applications; standard Spaghetti was the most versatile; Linguine and Fettuccine suited mixed, stir-fried, or broad-noodle styles. These culinary suggestions do not imply equivalence in ingredients or structure.



Figure 5. Original applications.

6. Discussion / 讨论

A memorable approximate 1.5-fold dimensional change emerged, but width and diameter are not equivalent physical measures and cannot support conclusions about cross-sectional area, volume, or water uptake.

7. Limitations / 局限性

Strand count, replication, measurement location, caliper pressure, and calibration were not documented. One author selected the sensory endpoint, and product formulation could influence results.

8. Conclusion / 结论

Under 4 g/L baking soda, recommended times ranged from 5 to 18 minutes and mean dimensional expansion was approximately 1.50. Future work should use repeated measurements, controlled caliper pressure, cooking-loss analysis, and instrumental texture testing.

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